

A Non-dominated Sorting Particle Swarm Optimizer for Multiobjective Optimization

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Abstract. This paper introduces a modified PSO, Non-dominated Sorting Particle Swarm Optimizer (NSPSO), for better multiobjective optimization. NSPSO extends the basic form of PSO by making a better use of particles' personal bests and offspring for more effective non-domination comparisons. Instead of a single comparison between a particle's personal best and its offspring, NSPSO compares all particles' personal bests and their offspring in the entire population. This proves to be effective in providing an appropriate selection pressure to propel the swarm population towards the Pareto-optimal front. By using the non-dominated sorting concept and two parameter-free niching methods, NSPSO and its variants have shown remarkable performance against a set of well-known difficult test functions (ZDT series). Our results and comparison with NSGA II show that NSPSO is highly competitive with existing evolutionary and PSO multiobjective algorithms.

1 Introduction

Multiobjective optimization problems represent an important class of real-world problems. Typically such problems involve trade-offs. For example, a car manufacturer may wish to maximize its profit, but meanwhile also to minimize its production cost. These objectives are typically conflicting to each other. A higher profit would increase the production cost. There is no single optimal solution. Often the manufacturer needs to consider many possible “trade-off” solutions before choosing the one that suits its need. The curve or surface (for more than 2 objectives) describing the optimal trade-off solutions between objectives is known as the Pareto front. One of the major goals in multiobjective optimization is to find a set of well distributed optimal solutions along the Pareto front.

In recent years, population-based optimization methods such as Evolutionary Algorithms have become increasingly popular for solving multiobjective optimization problems [1][2]. EA's success is due to its generic ability to handle large complex real-world problems. Since EAs maintain a population of solutions, this allows exploration of different parts of the Pareto front simultaneously. However not until recently, another population-based optimization

technique Particle Swarm Optimization (PSO) has been only applied to single objective optimization tasks. PSO technique is inspired by studies of social behavior of insects and animals [3]. The social behavior is modeled in a PSO to guide a population of particles (so-called swarm) moving towards the most promising area of the search space. In PSO, Each particle represents a candidate solution, $X_i = (x_{i1}, x_{i2}, \dots, x_{id})$. d is the dimension of the search space. The i -th particle of the swarm population knows: a) its *personal best position* $P_i = (p_{i1}, p_{i2}, \dots, p_{id})$, i.e., the best position this particle has visited so far that yields the highest fitness value; and b) the *global best position*, $P_g = (p_{g1}, p_{g2}, \dots, p_{gd})$, i.e., the position of the best particle that gives the best fitness value in the entire population; and c) its current *velocity*, $V_i = (v_{i1}, v_{i2}, \dots, v_{id})$, which represents its position change; The following equation (1) uses the above information to calculate the new updated velocity for each particle in the next iteration step. Equation (2) updates the each particle's position in the search space.

$$v_{id} = wv_{id} + c_1r_1(p_{id} - x_{id}) + c_2r_2(p_{gd} - x_{id}) \quad (1)$$

$$x_{id} = x_{id} + \chi v_{id} , \quad (2)$$

where $d = 1, 2, \dots, D$; $i = 1, 2, \dots, N$; N is the size of the swarm population; χ is a constriction factor which controls and constricts the velocity's magnitude; w is the inertia weight, which is often used as a parameter to control exploration/exploitation in the search space; c_1 and c_2 are two coefficients (positive constants); r_1 and r_2 are two random numbers within the range $[0, 1]$. There is also a V_{MAX} , which sets the upper and lower bound for velocity values.

PSO has proved to be an efficient optimization method for single objective optimization, and more recently has also shown promising results for solving multiobjective optimization problems [4][5] [6][7]. What is in common among these works is the use of a basic form of PSO first introduced by Kennedy and Eberhart [3]. However the basic form of PSO has some serious limitations in particular when dealing with multiobjective optimization problems. In PSO, a particle is modified only through its personal best and global best to produce its offspring. At each iteration step, if the fitness of the offspring is better than the parent's personal best, then the personal best is updated with this offspring, however, there is no sharing of information with other particles in the population, except that each particle can access the global best. For multiobjective optimization, we argue that such sharing of information among all the individuals in a population is crucial in order to introduce the necessary selection pressure to propel the population moving towards the true Pareto-optimal front.

This paper introduces a modified PSO, Non-dominated Sorting Particle Swarm Optimizer (NSPSO), which is able to increase such "sharing" among all particles in a swarm population especially concerning how to allow the population as a whole to progress towards the true Pareto-optimal front. For clarity, in this paper, we use P^* to denote the true Pareto-optimal front, and Q the found non-dominated solution set.